

# Assignment 2

(Questions from Module 3 & 4)

|                   |                        |
|-------------------|------------------------|
| <b>Name 1</b>     | NASAAIE BIN NORISKAMAR |
| <b>Matric No.</b> | A25CS0118              |
| <b>Section</b>    | 02                     |
| <b>Date</b>       | 15 MAY 2026            |

|                   |                            |
|-------------------|----------------------------|
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| <b>Matric No.</b> | A25CS0267                  |
| <b>Section</b>    | 02                         |
| <b>Date</b>       | 15 MAY 2026                |

Instructor Name : Dr Rashidah Binti Kadir

**Total Marks: 40**

## Instructions

- Students must answer **ALL questions** with complete workings, including conversions, arithmetic, and floating-point operations.
- Hardware algorithms should show full iteration tables with proper binary notation and indicate overflow, normalisation, truncation, or rounding.
- Complete this assignment in **pairs (2 members ONLY)** and **submit handwritten** work in **PDF format** through eLearning within 7 days, as specified by your instructor.

## Collaboration Requirements

Work together with your assigned partner and ensure both students contribute equally to the assignment. Communicate clearly and divide tasks fairly from the start.

## Submission Format

Write neatly by hand in black or blue ink on standard A4 paper, then scan and convert to PDF. Include your names, student ID numbers, course details, instructor name, and submission date on the cover page.

## Deadline and Submission

Confirm your specific 7-day deadline with your instructor. Submit only through the eLearning platform and keep a personal copy for your records.



# Question 1

|      |    |               |
|------|----|---------------|
| 0000 | 0A | myBytes       |
| 1    | 14 | myBytes + 1   |
| 2    | 1E | myBytes + 2   |
| 3    | 3E | } myWords     |
| 4    | 08 |               |
| 5    | F4 | } myWords + 1 |
| 6    | 01 |               |
| 7    | 00 | } myWords + 2 |
| 8    | 00 |               |
| 9    | 50 | } myWords     |
| A    | C3 |               |
| B    | 00 |               |
| C    | 00 | } myWords + 1 |
| D    | 10 |               |
| E    | 27 |               |
| F    | 00 | } myWords + 2 |
| 10   | 00 |               |
| 11   | 00 |               |
| 12   | 00 | } myWords + 2 |
| 13   | 00 |               |
| 14   | 00 |               |

|    |              |
|----|--------------|
| 00 | } setArr 1   |
| 69 |              |
| 00 | } setArr + 1 |
| 0B |              |
| 00 | } setArr + 2 |
| 6A |              |

|    |          |
|----|----------|
| 0F | setArr 2 |
| 00 | +1       |
| 5A | +2       |
| 5A | +3       |
| 43 | +4       |
| 41 | +5       |
| 52 | +6       |



## Question 1

- a)
- (i) ESI = 40000000h
  - (ii) EDI = 4000 0003h
  - (iii) CL = 0Ah
  - (iv) BX = 0014h
  - (v) EDX = 0001F400h
  - (vi) BX = 01F4h
- b)
- line 14 - Direct Addressing
  - line 18 - Direct Addressing
  - line 19 - Register Indirect Addressing
  - line 22 - Indexed Addressing

## a) Conversion part

set Arr 1 (Word - 2 bytes each)

- 3000 - 3001 : 105<sub>10</sub> → 0069h
- 3002 - 3003 : 1011<sub>10</sub> → 000Bh
- 3004 - 3005 : 6Ah → 006Ah

set Arr 2 (Byte - 1 byte each)

- 3006 = 15<sub>10</sub> → 0Fh
- 3007 = ? (Uninitialized)
- 3008 = 90<sub>10</sub> → 5Ah
- 3009 = 90<sub>10</sub> → 5Ah
- 300A = 'C' → 43h
- 300B = 'A' → 41h
- 300C = 'R' → 52h

## Question 2

a)

|      | Data (Hex) |           |
|------|------------|-----------|
| 3000 | 69         | Set Arr 1 |
| 3001 | 00         |           |
| 3002 | 0B         |           |
| 3003 | 00         |           |
| 3004 | 6A         |           |
| 3005 | 00         | Set Arr 2 |
| 3006 | 0F         |           |
| 3007 | ?          |           |
| 3008 | 5A         |           |
| 3009 | 5A         |           |
| 300A | 43         |           |
| 300B | 41         |           |
| 300C | 52         |           |

b)

- (i) AX = 000Bh
- (ii) DH = 5Ah
- (iii) BX = 6A00h
- (iv) DL = 43h

c) Yes, the program still can be assemble and run without error

Since the instruction of CALL Dumpregs is just a procedure from the library of Irvine32 that able to display the content in the register and flags to the screen.

The program still can be run without effect the logics or error. It just the user can not be able to see the value of output after the program end



### Question 3

$$Q = ((A+B) * C - D) / (E + F * G)$$

a) Three-address instructions

```
ADD T1, A, B
MUL T2, T1, C
SUB T3, T2, D
MUL T4, F, G
ADD T5, E, T4
DIV Q, T3, T5
```

(6 lines)

d) NO address instructions

```
PUSH A
PUSH B
ADD
PUSH C
MUL
PUSH D
SUB
PUSH E
PUSH F
PUSH G
MUL
ADD
DIV
POP Q
```

(14 lines)

b) Two-address instructions

```
MOV R1, A
ADD R1, B
MUL R1, C
SUB R1, D

MOV R2, F
MUL R2, G
ADD R2, E

DIV R1, R2
MOV Q, R1
```

(9 lines)

c) One address instructions

```
LOAD A
ADD B
MUL C
SUB D
STORE T1

LOAD F
MUL G
ADD E
STORE T2

LOAD T1
DIV T2
STORE Q
```

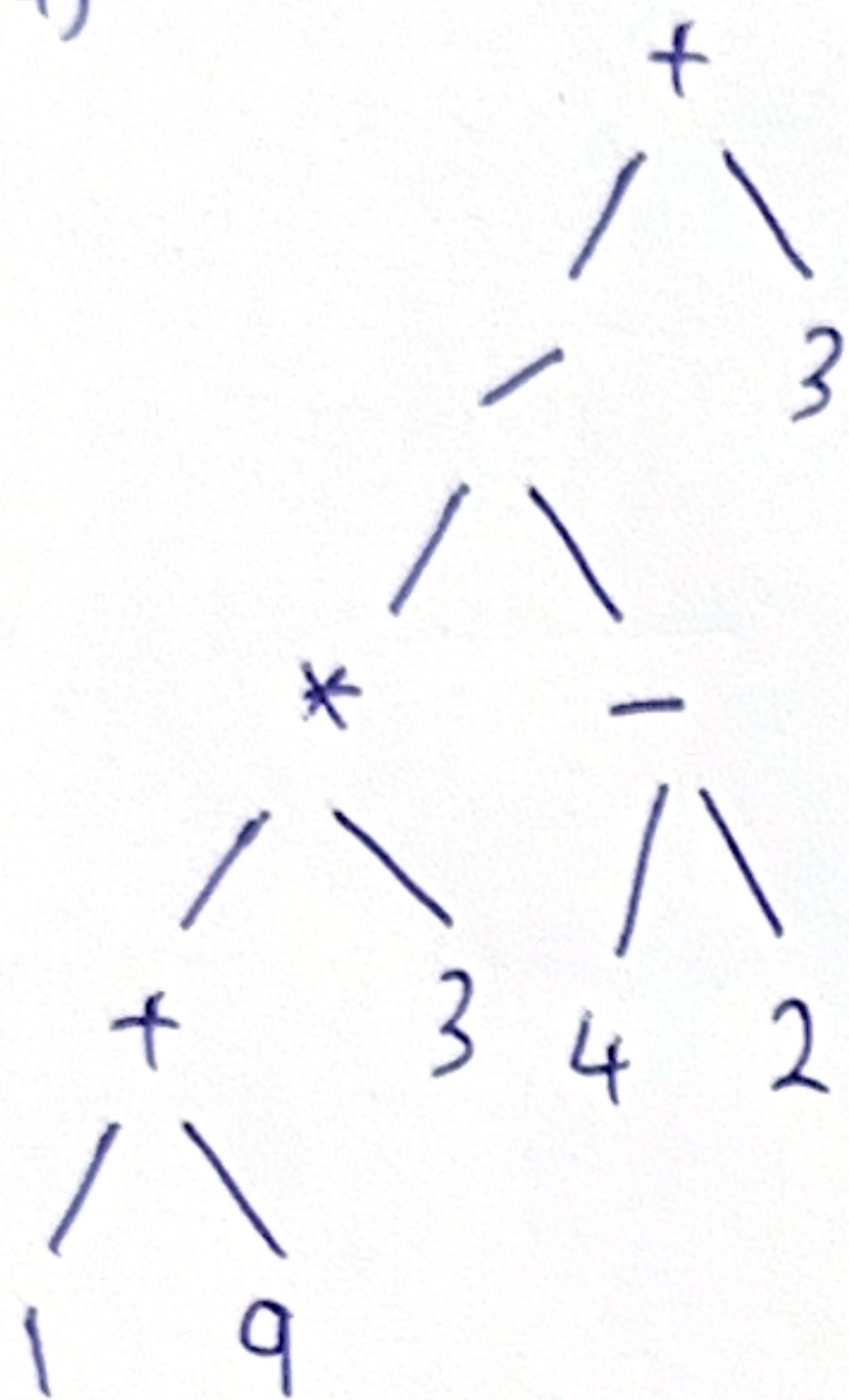
(12 lines)



## Question 4

$$M = ((1+9) * 3) / (4-2) + 3$$

a)



b)  $(1+9) \rightarrow 19+$

$((1+9)*3) \rightarrow 19+3*$

$(4-2) \rightarrow 42-$

$(((1+9)*3)/(4-2)) \rightarrow 19+3*42- /$

$\therefore 19+3*42- / 3+$

$1+9 = 10$

$10*3 = 30$

$4-2 = 2$

$30/2 = 15$

$15+3 = 18$

$\therefore M = 18$

c) i) One-address instructions

LOAD 4

SUB 2

STORE T1

LOAD 1

ADD 9

MUL 3

DIV T1

ADD 3

STORE M

(9 lines)

ii)

PUSH 1

PUSH 9

ADD

PUSH 3

MUL

PUSH 4

PUSH 2

SUB

DIV

PUSH 3

ADD

POP M

(12 lines)